

GaussNiche 5-D downstream HSM — full report (RND / buffer-out / uniform+)

Median per metric x sampler — pc5 predictors

metric	RND	buffer-out	uniform+
COR (truth)	0.759	0.725	0.638
RMSE (truth)	0.222	0.275	0.310
AUC	0.898	0.970	0.969
CBI	0.842	0.822	0.798
TSS	0.722	0.867	0.867
Sensitivity	0.933	0.956	0.967
Specificity	0.811	0.929	0.922
Kappa	0.722	0.867	0.867
COR	0.718	0.864	0.866
R2	0.526	0.750	0.755
RMSE	0.354	0.271	0.273
Deviance	135.359	82.380	80.932

Median per metric x sampler — raw12 predictors

metric	RND	buffer-out	uniform+
COR (truth)	0.730	0.687	0.574
RMSE (truth)	0.237	0.297	0.354
AUC	0.887	0.964	0.962
CBI	0.849	0.832	0.790
TSS	0.700	0.856	0.856
Sensitivity	0.931	0.944	0.956
Specificity	0.800	0.922	0.911
Kappa	0.700	0.856	0.856
COR	0.700	0.851	0.850
R2	0.522	0.746	0.752
RMSE	0.366	0.284	0.286
Deviance	145.421	91.746	90.848

Internal metrics = held-out test vs sampled PA; *_truth = predicted map vs known suitability.

Where uniform+ beats the naive samplers — pc5 predictors

cell = margin of uniform+ vs the BETTER naive sampler (>0 better); median over algos x realisations

	Generalist · common	Specialist · common	Generalist · rare	Specialist · rare
COR (truth)	-0.065	-0.088	-0.141	-0.033
RMSE (truth)	-0.100	-0.080	-0.102	-0.017
AUC	-0.000	-0.001	-0.014	-0.004
CBI	-0.073	-0.046	-0.028	+0.003
TSS	+0.006	+0.011	-0.022	+0.008
Sensitivity	+0.022	+0.011	+0.022	+0.000
Specificity	-0.011	-0.011	-0.044	+0.000
Kappa	+0.006	+0.011	-0.022	+0.008
COR	+0.005	+0.001	-0.021	+0.000
R2	+0.018	-0.003	-0.026	-0.040
RMSE	+0.003	+0.006	-0.011	-0.005
Deviance	+1.727	-0.388	-7.087	-0.797

uniform+ vs naive
(RND & buffer-out)

beats both

beats one

beats none

Where uniform+ beats the naive samplers — raw12 predictors

cell = margin of uniform+ vs the BETTER naive sampler (>0 better); median over algos x realisations

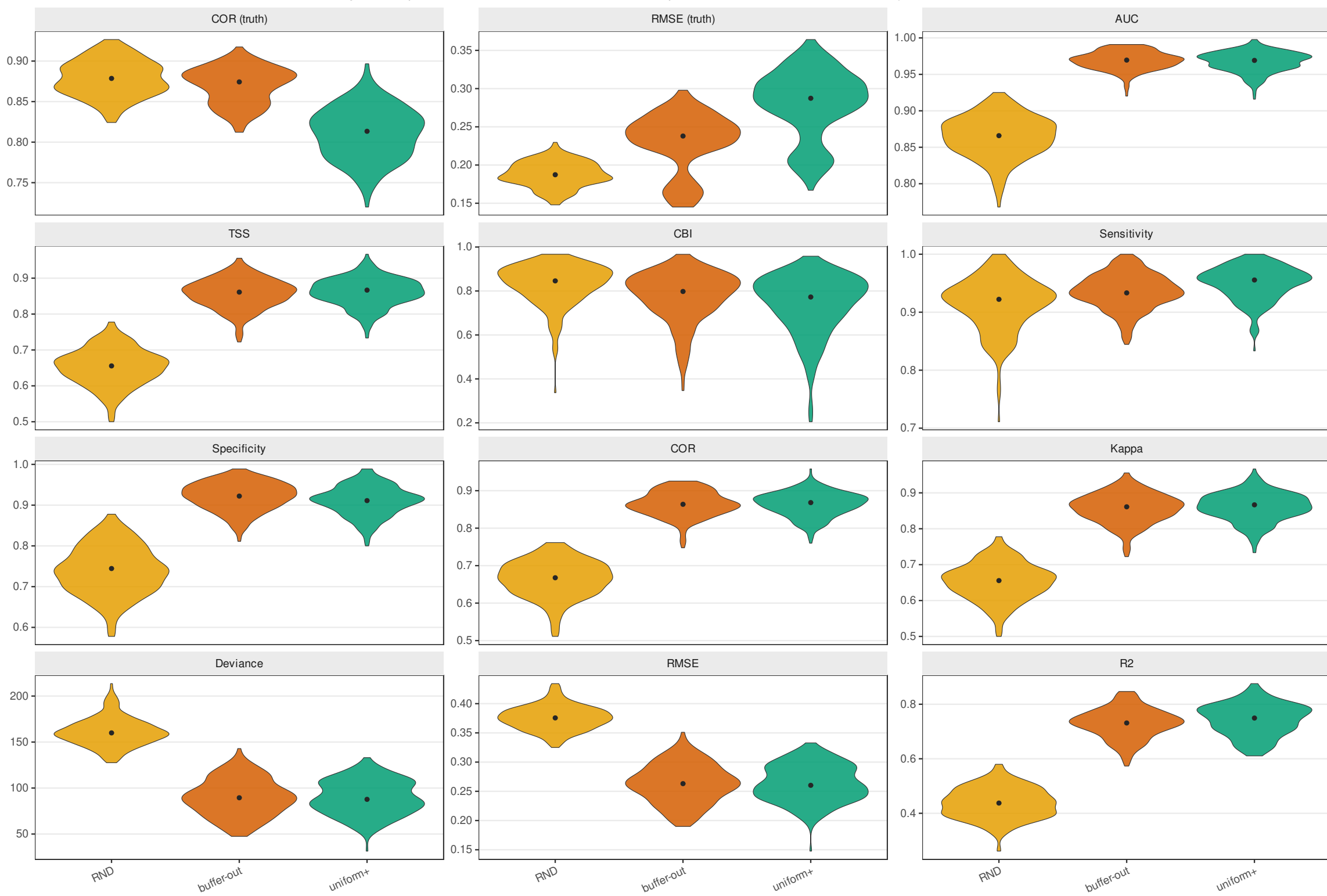
COR (truth)	-0.106	-0.124	-0.155	-0.052
RMSE (truth)	-0.128	-0.102	-0.110	-0.041
AUC	-0.001	-0.004	-0.016	-0.011
CBI	-0.121	-0.076	-0.002	-0.056
TSS	+0.011	+0.000	-0.022	-0.033
Sensitivity	+0.022	+0.000	+0.000	-0.028
Specificity	-0.011	-0.006	-0.033	-0.044
Kappa	+0.011	+0.000	-0.022	-0.033
COR	+0.008	-0.001	-0.030	-0.017
R2	+0.027	-0.005	-0.043	-0.048
RMSE	+0.004	+0.001	-0.014	-0.006
Deviance	+1.866	-7.222	-8.930	-0.217
	Generalist · common	Specialist · common	Generalist · rare	Specialist · rare

uniform+ vs naive
(RND & buffer-out)

beats both
beats one
beats none

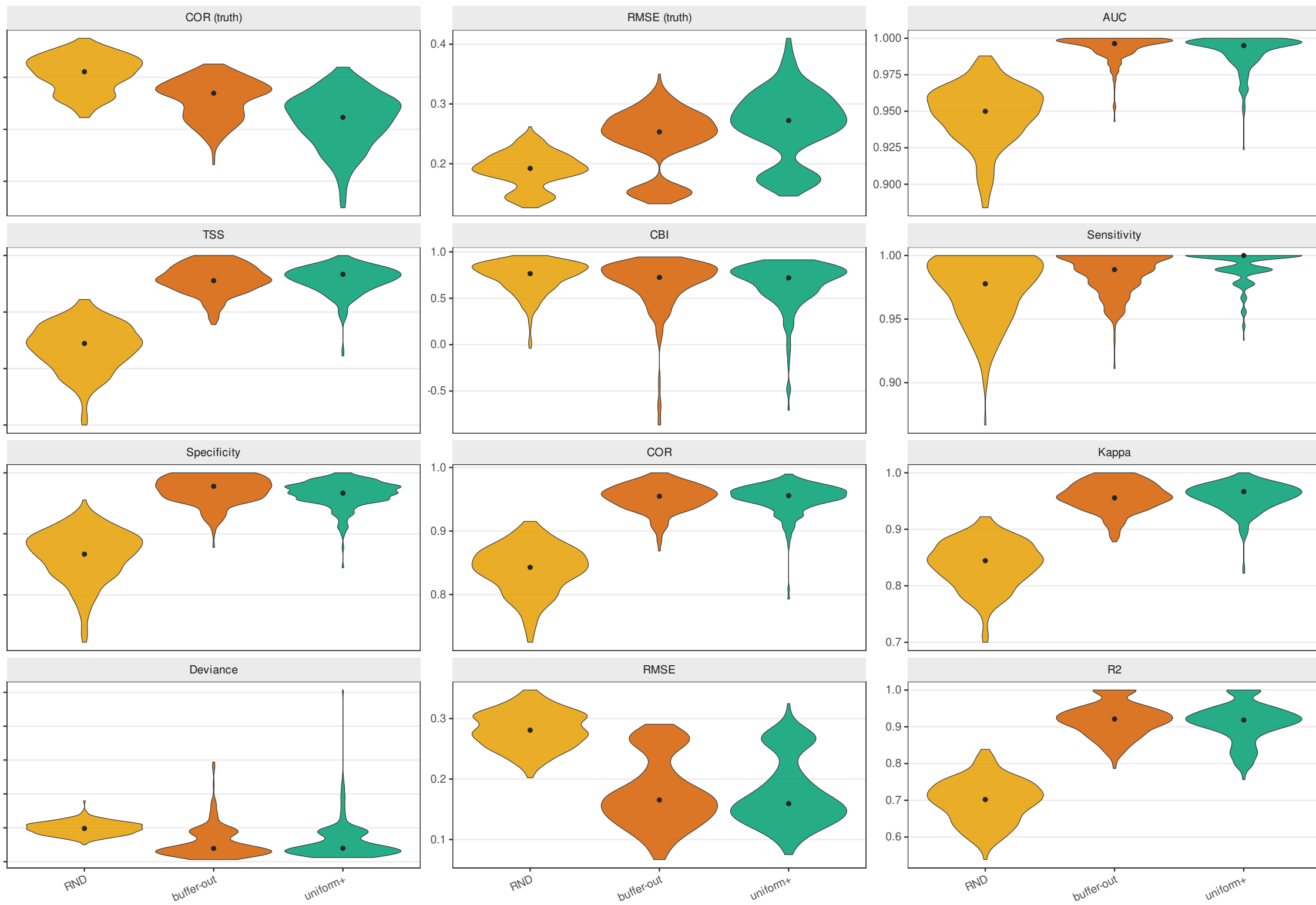
Generalist · common — pc5 predictors

violins: distribution over 50 realisations x 5 HSM algorithms; point = median. Internal metrics = vs sampled PA; *_truth = vs known suitability



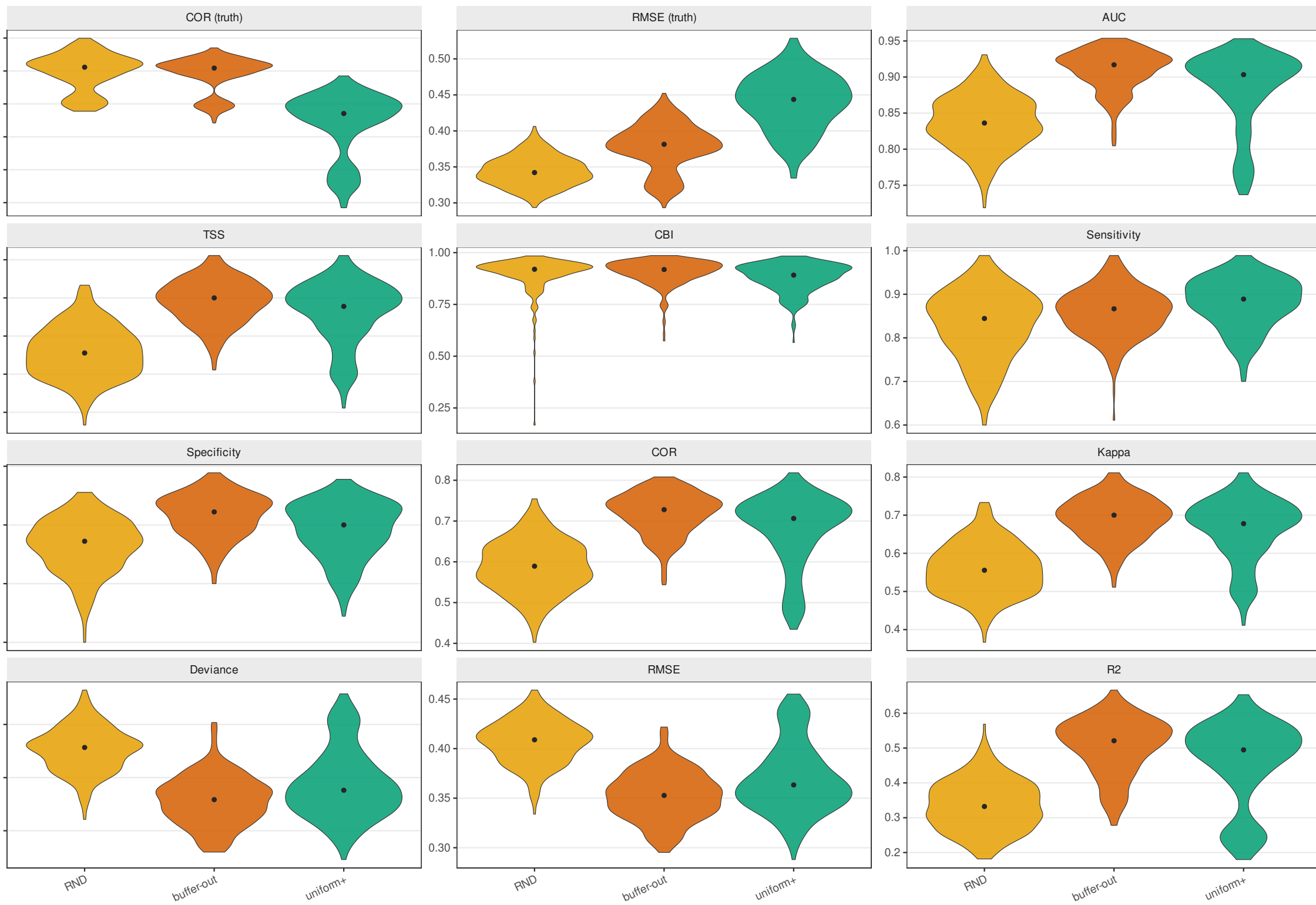
Specialist · common — pc5 predictors

violins: distribution over 50 realisations x 5 HSM algorithms; point = median. Internal metrics = vs sampled PA; *_truth = vs known suitability



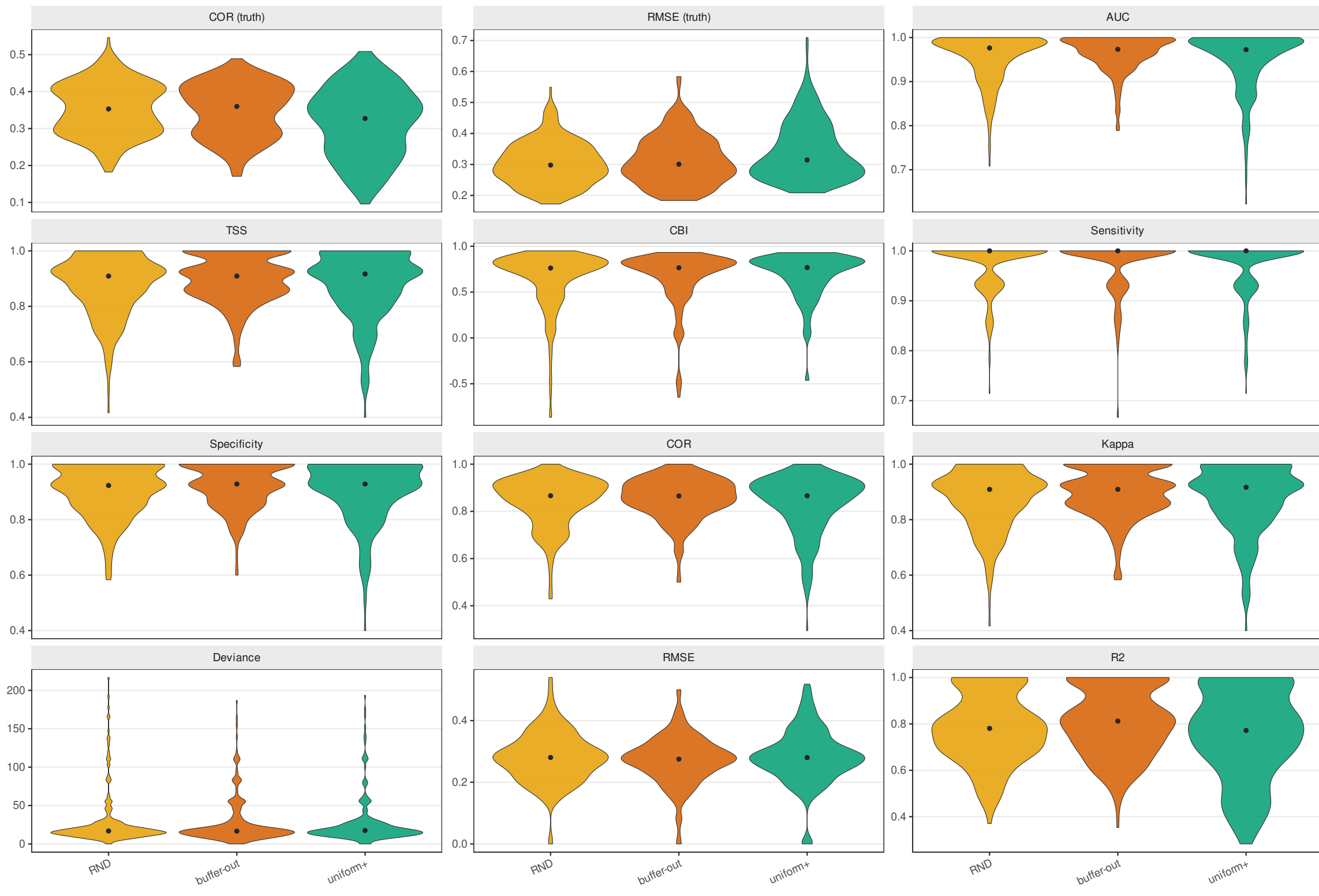
Generalist · rare — pc5 predictors

violins: distribution over 50 realisations x 5 HSM algorithms; point = median. Internal metrics = vs sampled PA; *_truth = vs known suitability



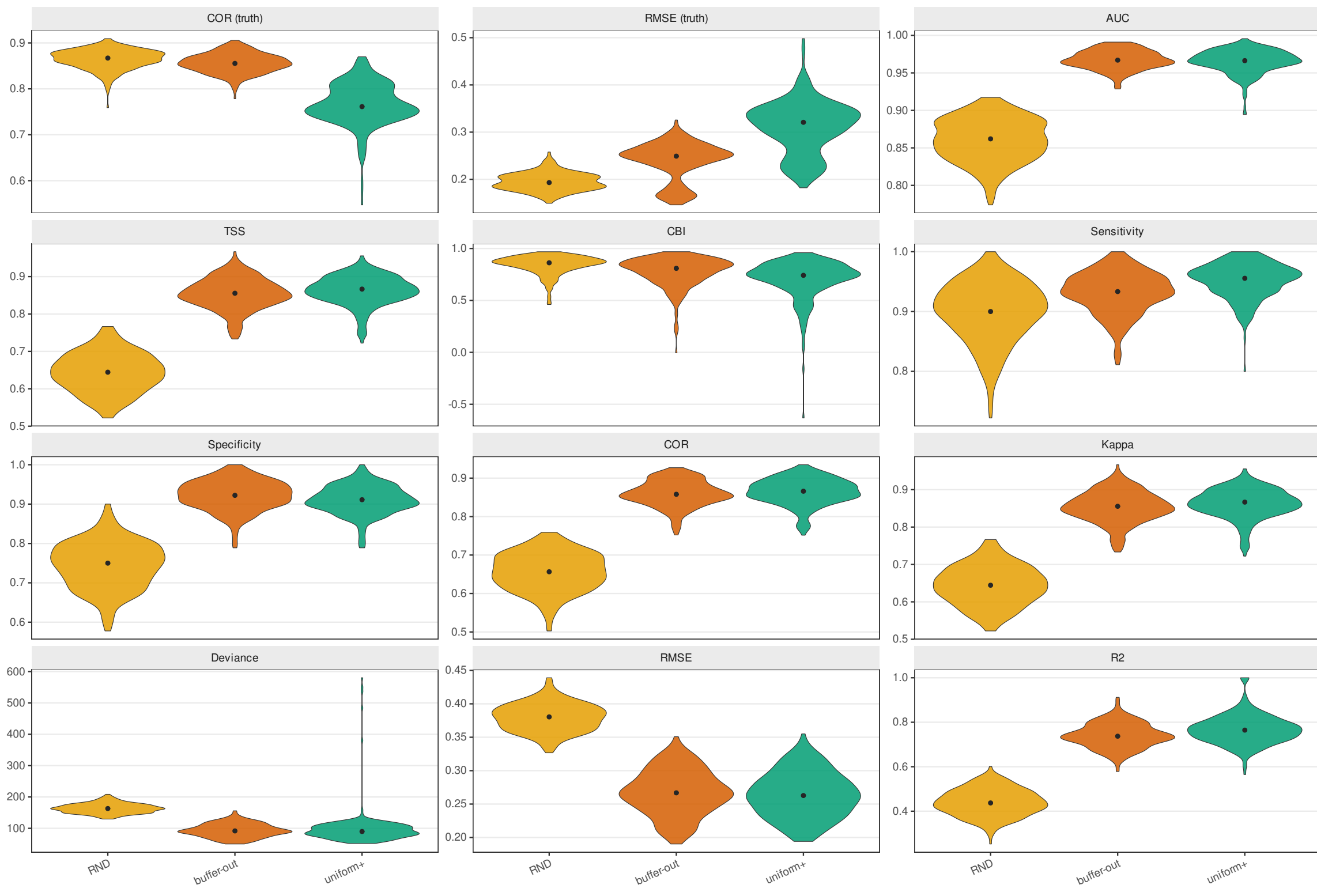
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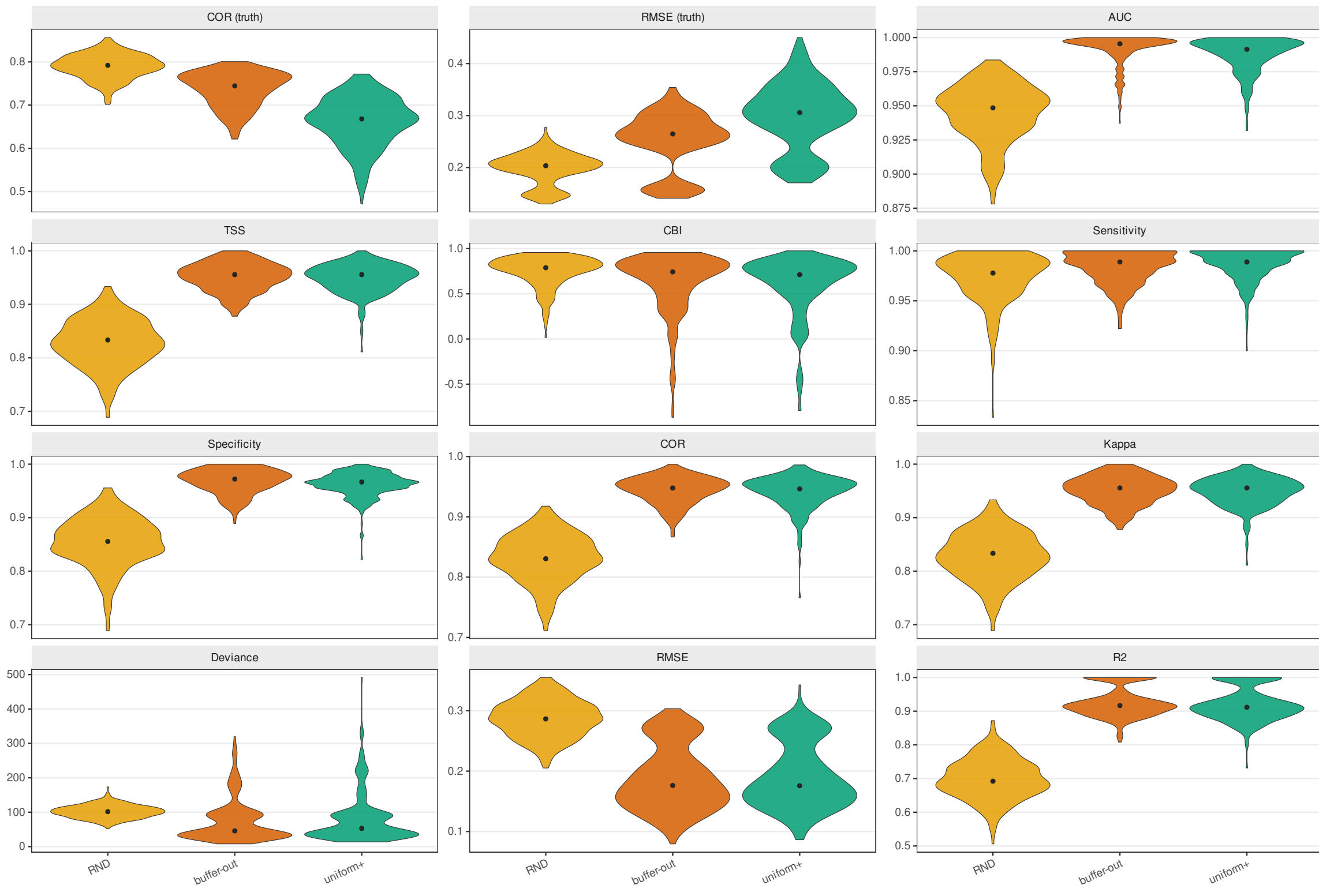
Generalist · common — raw12 predictors

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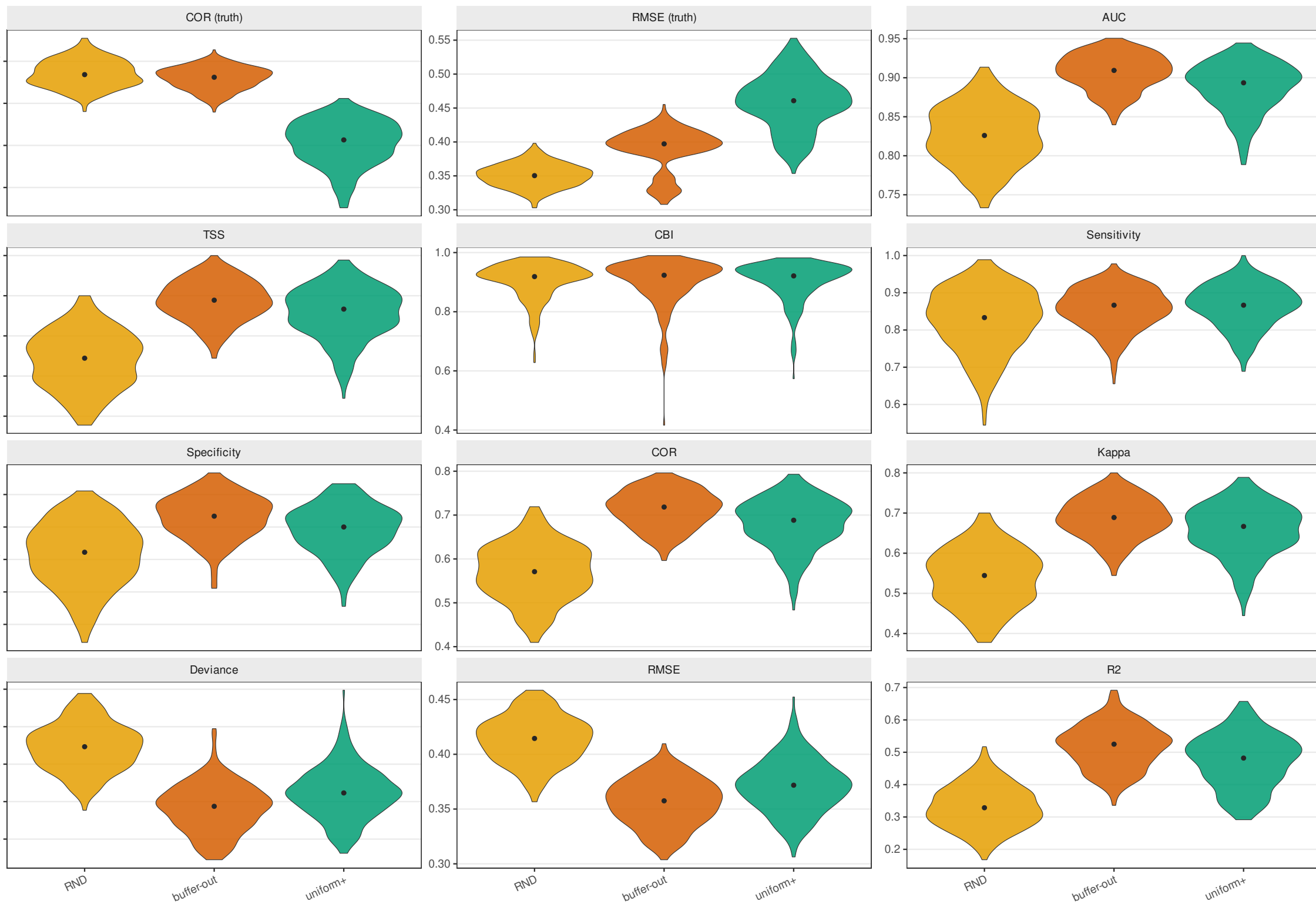
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